

Forest and Landscape Structure as Predictors of Capercaillie Occurrence

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ABSTRACT Capercaillie (*Tetrao urogallus*) is a large, endangered forest grouse species with narrow habitat preferences and large spatial requirements that make it susceptible to habitat changes at different spatial scales. Our aim was to evaluate the relative power of variables relating to forest versus landscape structure in predicting capercaillie occurrence at different spatial scales. We investigated capercaillie–habitat relationships at the scales of forest stand and forest-stand mosaic in 2 Swiss regions. We assessed forest structure from aerial photographs in 52 study plots each 5 km². We classified plots into one of 3 categories denoting the observed local population trend (stable, declining, extinct), and we compared forest structure between categories. At the stand scale, we used presence–absence data for grid cells within the plots to build predictive habitat models based on logistic regression. At this scale, habitat models that included only variables relating to forest structure explained the occurrence of capercaillie only in part, whereas variables selected by the models differed between regions. Including variables relating to landscape features improved the models significantly. At the scale of stand mosaic, variables describing forest structure (e.g., mean canopy cover, proportion of open forest, and proportion of multistoried forest) differed between plot categories. We conclude that small-scale forest structure has limited power to predict capercaillie occurrence at the stand scale, but that it explains well at the scale of the stand mosaic. Including variables for landscape structure improves predictions at the forest-stand scale. Habitat models built with data from one region cannot be expected to predict the species occurrence in other regions well. Thus, multiscale approaches are necessary to better understand species–habitat relationships. Our results can help regional authorities and forest-management planners to identify areas where suitable habitat for capercaillie is not available in the required proportion and, thus, where management actions are needed to improve habitat suitability. (JOURNAL OF WILDLIFE MANAGEMENT 71(2):356–365; 2007)

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The role of scale in ecology has been discussed extensively over the past 2 decades and is now widely recognized as being of vital importance for understanding ecological processes (Wiens 1989, Levin 1992). As species–habitat interactions usually involve several spatial scales, habitat analyses should adopt multiscale approaches for better understanding patterns of species occurrence (Addicott et al. 1987, Wiens 1989, Bissonette 1997, Storch 1997). There has been fast progress in remote sensing, Geographic Information Systems, and statistical analysis tools that initiated a large number of multiscale studies (e.g., Fuhlendorf et al. 2002, Lawler and Edwards, Jr. 2002, Zabel et al. 2003, Fischer et al. 2004, Graf et al. 2005). Such studies often use habitat data of low resolution (e.g., Mladenoff and Sickley 1998, Gibson et al. 2004) and may, thus, have to work with indirect or surrogate variables (Mladenoff et al. 1997). However, efficient conservation and management of endangered species requires knowledge of those habitat factors that causally determine habitat suitability and, hence, the occurrence of local populations. Thus, larger-scale modeling approaches should include habitat predictors to which the species directly responds (e.g., Thompson and McGarigal 2002). Yet, assessing

important habitat features of species with large home ranges will often require huge resources in terms of personnel and funding. As a compromise, habitat suitability models at the scale of home ranges or local populations may be based on variables that can be measured easily over large areas (e.g., from remote-sensing data with high resolution) as long as they are of sufficiently direct meaning to the species in question.

Aerial photographs can be used for assessing the structure of and changes in forests and landscapes (D'Aoust et al. 2004, Holder 2004) and, thus, for investigating present and past habitat quality (e.g., Kadmon and Harari-Kremer 1999, McGrath et al. 2003, Marchand and Litvaitis 2004). Even specific habitat features such as the presence of spruce (*Picea abies*) snags (Bütler and Schlaepfer 2004) or stand characteristics indicating lichen diversity (Ask and Nilsson 2004) can be quantified from color infrared aerial photographs.

Capercaillie (*Tetrao urogallus*) is a large forest grouse species with specialized habitat preferences and extensive spatial requirements (Rolstad 1988, Storch 1995), a combination that makes it highly susceptible to habitat and landscape changes. Capercaillie populations are declining in most of their central European range (Storch 2000), as loss and fragmentation of suitable habitat have split populations into smaller units that are only loosely connected or even completely isolated from each other (e.g., Klaus 1991). In Switzerland, the downward trend is

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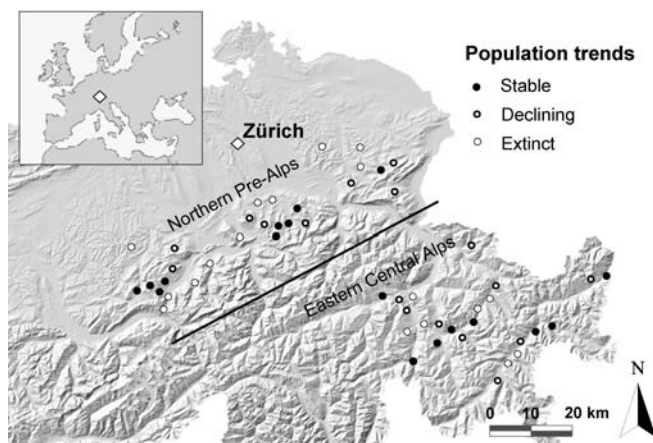


Figure 1. Locations of the 52 study plots (5 km² each) in the Swiss Alps where we studied capercaillie–habitat relationships, 1995–2003. The plots belong to 3 categories: stable, declining, and extinct, according to capercaillie population status derived from the 3 national capercaillie inventories of 1971, 1986, and 2001 (Mollet et al. 2003). The solid line separates the regions of the northern Pre-Alps and the eastern Central Alps. In the 52 plots, we assessed: 1) capercaillie presence–absence pattern (own field surveys, records of game wardens and local experts 1995–2003), 2) forest structure from black-and-white aerial photographs (1993–2000), and 3) landscape structure from geographical data with country-wide coverage (national data bases, state 2001).

particularly pronounced, and capercaillie faces a high extinction risk as only some 900–1,000 individuals remain in highly fragmented populations (Mollet et al. 2003).

Many studies at the local scale have elucidated habitat requirements of capercaillie. In the central European and Scandinavian populations, important habitat features are intermediate canopy cover (40–70%; Gjerde 1991b, Storch 1993b, Suchant 2002) and rich ground vegetation cover (e.g., Klaus et al. 1985, Picozzi et al. 1992, Sjöberg 1996, Bollmann et al. 2005) that is ideally dominated by bilberry (*Vaccinium myrtillus*; Eiberle 1976, Schroth 1992, Storch 1993a). Telemetry studies have revealed that spatial requirements of capercaillie are extensive (home range size 100–1,000 ha; e.g., Rolstad 1988, Storch 1995) and that populations are sensitive to macrohabitat alterations such as forest fragmentation or the spatial arrangement of favorable habitat (Rolstad and Wegge 1987, 1989). Old-growth forest makes good habitat quality, and with increasing proportion of old-growth stands the number of lekking males increases while home range size decreases (Wegge and Rolstad 1986, Storch 1995). Recent population studies on capercaillie ecology indicate that reproductive success is at least partly explained by landscape factors operating at a scale that is an order of magnitude larger (approx. 100 km²) than the area used by an individual grouse (a few square kilometers, Kurki et al. 2000). Still, for the efficient conservation of capercaillie or other grouse species, habitat analyses at multiple spatial scales are necessary, especially in regions where the species are endangered (Storch 2002, Keppie and Kierstead 2003).

The goal of this study was to investigate capercaillie–habitat relationships at the 2 scales of forest stands and forest-stand mosaics. We analyzed the power and limitation

of forest-structure variables for predicting capercaillie occurrence at the 2 scales and compared their predictive power with that of a set of landscape variables. To investigate how capercaillie–habitat relationships vary with spatial extent, we built separate models for 2 ecologically differing regions and tested all models on independent data.

STUDY AREA

The distribution of capercaillie in Switzerland is split into 5 independent meta-populations across 3 mountain regions: one in the Jura mountains, 3 in the northern Pre-Alps, and one in the eastern Central Alps (Mollet et al. 2003). We used data from the northern Pre-Alps and the eastern Central Alps (Fig. 1). These 2 geographic regions differed considerably in terms of climate, topography and geology, forest distribution, and human land use. The climate of the eastern Central Alps (600–3,500 m above sea level [asl]) was continental with a relatively low precipitation rate (800–2,000 mm/yr) and cold winters but warm and dry summers. Large contiguous and mostly conifer-dominated forests were spread along the valley slopes, and the natural tree line was at an elevation of approximately 2,300 m. Generally, human population density and impact by tourism were low, except for some vacation centers with both summer and winter tourism.

In the northern Pre-Alps (400–2,800 m asl), the climate was more atlantic with cold-temperate winters and wet summer months that held the upper tree-line at about 2,000 m asl. Because of the high precipitation (2,000–3,000 mm/yr), forests were interspersed with mires where impervious soils and topography allowed. Here, forest stands were naturally open and diversely structured and had low timber value. Such areas are favored by capercaillie and other woodland grouse species. Northern Pre-Alpine landscapes have been altered by long-term livestock farming, and there was a patchy pattern of forests and pastures over large areas as a result. The northern Pre-Alps were within half a day's drive of major Swiss cities and, thus, frequently used for recreation, with the resulting disturbance probably affecting the survival rate and reproductive success of capercaillie.

METHODS

Study Plots and Species Data

We used 52 study plots (circles) of 5 km² each for the analyses at the scale of forest-stand mosaics. This plot size corresponds to the average home range size measured in telemetry studies in a Central European region (Storch 1995). We chose the locations of the study plots by a 2-step procedure: First, we classified potential capercaillie areas (forest cover ≥50% within 5 km²) by their population trend as stable, declining, or extinct. We took the necessary data from the 3 national capercaillie surveys (1971, 1985, and 2001; Mollet et al. 2003) in which local populations were counted using the same method (lek counts of displaying males). We classified areas as stable when local population(s) did not decline by more than 40% between 1971 and 2001, and we classified areas as declining when population(s)

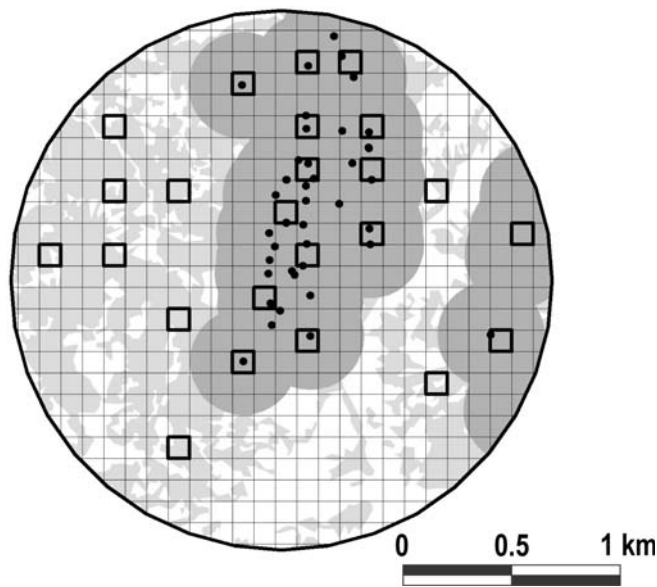


Figure 2. Coding of capercaillie presence-absence within our study plots in the Swiss Alps, 1995–2003, here shown on the basis of a 1-ha grid for better illustration (in reality, we used a grid with a cell size of 400 m², see Methods). Black points: capercaillie observations; dark grey area: buffer of 250 m around observations; light grey: forested areas without capercaillie observations. We defined grid cells outside the buffer area as absence. The subset of cells we used in the analyses is highlighted in bold.

decreased clearly (>40%). We classified areas that had a local population in 1971 or 1985 but were not occupied in 2001 as extinct. There was no category increasing because no population increased by >40% in the period between 1971 and 2001. Second, we randomly selected the locations of 52 study plots so that: 1) plots were located at minimum distances of 1.3 km (radius of study plots; edge-to-edge distance) from each other and 2) we equally sampled the 3 plot categories as well as the 2 regions (Pre-Alps and Alps). Of the 52 plots, 28 were located in the Pre-Alps (category stable: 9 plots, declining: 8 plots, extinct: 11 plots) and 24 in the Alps (stable: 8, declining: 9, extinct: 7; Fig. 1).

Capercaillie did not necessarily use the entire area of a study plot. Therefore, we assumed that those parts of the study plots with actual capercaillie observations were used regularly by capercaillie, and that the other parts were, if at all, used only temporarily by the species. Thus, we assumed areas with capercaillie observations to be suitable habitat. On the other hand, areas without observations were not necessarily unsuitable, but they probably were on average less appropriate for the species.

We used a comprehensive set of observational data from different sources (own field surveys, records of game wardens and local experts from the time period 1995–2003) to identify presence and absence areas within the plots (Fig. 2) as a basis for calibrating the regression models. We assigned the observation data to a grid with a cell size of 20 m. From these grid cells, we extracted a sample of 296 cells from the Pre-Alps (presence: 146, absence: 150) and 180 from the Central Alps (presence: 90, absence: 90). We randomly sampled presence cells, but there had to be ≥ 150 m between presence cells. We randomly chose absence cells

from a regular 250-m net, and there had to be ≥ 250 m between presence and absence cells.

Our results will primarily apply to winter habitat selection, as our capercaillie occurrence data stemmed mostly from winter surveys. In the Pre-Alps, there was good evidence that the winter ranges also provided preferred summer habitat (R. Hess and F. Rudmann, Intercantonal Task Group for Grouse, personal communication). For the Central Alps, however, such information was not available and winter and summer ranges may in fact differ locally, as observed in other parts of the capercaillie distribution range (Rolstad and Wegge 1989, Storch 1995). At the scale of entire study plots (5 km²), we assumed that we included both winter and summer ranges.

Assessing Forest Structure from Aerial Photographs

We assessed forest-structure variables from black-and-white aerial photographs with a ground resolution of 65 cm (aerial photographs © swisstopo, DVO033373; 1993–2000). We ortho-rectified the images by monoplotting in Erdas Imagine 8.6 (Leica Geospatial Imaging, LLC, Norcross, GA), and we used a digital elevation model with 25-m resolution as vertical reference file (DHM25 © 2004, swisstopo, DVO033594). Ortho-rectification of the images was very exact (± 5 m) because image orientation parameters were available.

We developed a standardized manual interpretation procedure to assess forest-structure variables from the ortho-rectified aerial photographs. We delineated (digitized) polygons with uniform forest stands or clearings (gaps) in ArcView3.2a in compliance with a specified set of rules: 1) fixed view-scale at 1:5,000 and 2) minimum size of forest patches of 0.25 ha. We then estimated a set of structural variables for each forested polygon. We trained patch delineation and forest-structure assessment in test areas and verified these results in the field.

We chose variables that are known to be important for capercaillie habitat quality and that are recognizable in aerial photographs: succession state, overall canopy cover, tree cover, understorey cover, vertical structure, and horizontal structure (Table 1). Most of these variables were identical to or had been adapted from variables assessed in the Swiss National Forest Inventory (Brassel and Brändli 1999, Brassel and Lischke 2001).

The manual interpretation of aerial photographs entails the problem of accuracy and reproducibility of the data. We tested the quality of our data with 2 approaches. First, we assessed variables describing forest structure in the field (ground-truthing) in 179 patches from 4 study plots (2 each in Pre-Alps and Alps). We successfully assessed estimates of successional state, horizontal and vertical stand structure, and overall canopy cover (>50% of the patches were identically classified, and over 95% were assigned to at least the neighboring category), whereas the understorey cover was difficult to estimate from aerial photographs (only about 40% of the polygons classified identically, about 80% assigned to at least neighboring category). Second, we assessed a part of the data twice to measure reproducibility

Table 1. Description of forest-structure variables that we used to analyze capercaillie–habitat relationships at 2 spatial scales (forest stand, forest-stand mosaic) in 2 Swiss regions (Pre-Alps, Alps). We identified these variables from black-and-white aerial photographs (1993–2000) with a ground resolution of 0.65 m.

Variable description	Abbreviation	Unit	Dropped because of correlation with ($r_s > 0.7$)	
			Pre-Alps	Alps
Total canopy cover; 1: open area, 2: canopy cover of 1–10%, 3: 10–25%, 4: 25–50%, 5: 50–75%, 6: 75–90%, 7: 90–100%	CC	Cat	CC100	CC100
$\bar{x}$ of total canopy cover within a 100-m circle around the center of the grid cell	CC100	%		
Canopy cover of tree layer (trees with dbh > 12 cm); 1: open area, 2: canopy cover of 1–10%, 3: 10–25%, 4: 25–50%, 5: 50–75%, 6: 75–90%, 7: 90–100%	CCT	Cat	CC100	CC100
$\bar{x}$ of tree cover within a 100-m circle around the center of the grid cell	CCT100	%	CC100	CC100
Understory cover (bushes and trees with dbh ≤ 12 cm); 1: open area, 2: cover of 1–10%, 3: 10–25%, 4: 25–50%, 5: 50–75%, 6: 75–90%, 7: 90–100%	CCU	Cat		
$\bar{x}$ of understory cover within a 100-m circle around the center of the grid cell	CCU100	%		
Succession state; 1: young-growth stand, 2: pole stand, 3: old-growth, 4: mixed	SS	Cat		
Horizontal stand structure; 1: no clusters, 2: open clusters, 3: dense clusters (overlapping tree-crowns)	HS	Cat	CC100	CC100
Sum of clustered forests (HS > 1) within a 100-m circle around the center of the grid cell	HS100	ha	CC100	CC100
Vertical stand structure; 1: 1-layered, 2: 2-layered, 3: multilayered	VS	Cat	VS100	
Sum of multistoried forests (VS > 1) within a 100-m circle around the center of the grid cell	VS100	ha		
Length of forest edges; sum of length of forest edges to open area	FE100	m		
Variation of canopy cover; no. of different canopy-cover categories occurring within a 100-m circle around the center of the grid cells	CCV100	index		
Variation of succession state; no. of different succession-state categories occurring within a 100-m circle around the center of the grid cells	SSV100	index		FE100

of the interpretation. In 82 quadratic cells of 0.25 km² (40 distributed over 10 study plots in the Pre-Alps, 42 within 12 study plots in the Alps) we repeated data assessment and compared mean values per cell between the 2 assessments using bivariate correlation (Pearson). The distinction between forest and open area was most reliable (Pearson Correlation = 0.98), and the other variables also attained a correlation of at least 0.80.

In the forest-structure variables, we included both the values at the exact location of a grid cell and the mean values or sums from a circular analysis window with a radius of 100 m (corresponding to the daily activity range of an individual capercaillie; Table 1). For instance, dense forests with a poor field layer are usually assumed to be unsuitable for capercaillie (Schroth 1992, Storch 1993b, Suchant 2002). However, a dense group of trees surrounded by open forest or close to a gap or opening can provide shelter for capercaillie. We captured such mosaics at least to some degree with the moving window approach.

Landscape-Scale Variables

We compared the predictive power of the forest-structure variables described above with the power of landscape variables that are assumed to be relevant for capercaillie occurrence and that we had used before (Table 2; see also Graf 2005, Graf et al. 2005). We included these variables in the analysis at different spatial scales by calculating the

mean, sum, or majority values for different moving windows from 1 ha to just over 1,100 ha (ArcInfo 9.0, focal statistics). We used the variables at the scales at which they were found to explain the occurrence of capercaillie best in univariate analyses (Graf 2005). That is, we chose the best explaining scale for every variable and for the 2 regions (Pre-Alps, Alps) separately.

Statistical Analyses

At the scale of entire study plots, we compared means or sums of variables between the plot categories. First, we compared presence (categories stable and declining pooled) against absence (extinct). Then, within the presence data we compared the population trends stable and declining. We applied nonparametric test statistics (i.e., Mann–Whitney *U*-test [α -level 0.05]) to identify significant differences.

At the forest-stand scale (cell scale), we built 4 separate habitat suitability models using logistic regression (see below) and data from the Pre-Alps and the Alps, and we included either variables describing only forest structure or variables for both forest and landscape structure. We then applied all 4 models to the entire study plots. We defined grid cells predicted as presence as having a predicted probability of capercaillie occurrence higher than 0.5. We then compared the proportion of grid cells predicted as presence among plot categories using Kruskal–Wallis and Mann–Whitney tests.

Table 2. Landscape variables originating from geographical data with country-wide coverage (national data bases, state 2001) that we used in earlier analyses at larger spatial scales (Graf et al. 2005). We compared the power of these variables to predict capercaillie occurrence in 2 Swiss regions, 1995–2003, with that of forest-structure variables (Table 1).

Variable description	Abbreviation	Unit	Spatial grain size used for modeling (ha)	
			Pre-Alps	Alps
Slope	SLOPE	degrees	5	5
Topographic position	TOP	unitless	1	29
$\bar{x}$ temp (Jun)	TAVE	°C $\times$ 0.01	1	253
Proportion of forest	PFOR	% $\times$ 4	253	81
Proportion of mires and wet forests	MIRE	% $\times$ 4	253	5
Density of motorable roads	ROD	m/ha	113	49
Proportion of settlements	SETTL	0/1	529	81

Logistic regression.—We applied logistic regression (Manly et al. 2002, Menard 2002) for all habitat modeling using the software SPSS 11.0 (SPSS, Inc., Chicago, IL). Following Hosmer and Lemeshow (2000), we used a binomial error distribution and a logit link function. In all modeling, we included untransformed variables, as normality is not required and error terms are allowed to have non-Gaussian distributions (Guisan and Zimmermann 2000). We evaluated the type of response by plotting the frequency distribution of the predictor variables for both presence and absence plots. In the case of a unimodal response, we included the squared predictor variables in the univariate and multivariate analyses as well (Guisan and Zimmermann 2000).

Variable selection.—Three procedures helped us in deciding whether to include or exclude a predictor variable in the multivariate regression models. First, in pairs of highly correlated variables (Spearman's rank correlation higher than 0.7; Fielding and Haworth 1995), we excluded the one with no (or less) direct influence on capercaillie populations as judged on biological grounds (Table 1). In landscape variables, we excluded the density of motorable roads due to high correlation with average temperature. Second, the algebraic sign of the coefficient of a variable had to be ecologically plausible. Otherwise, we omitted the variable. Third, in the stepwise model-building procedure we used a threshold of $P < 0.1$ for the decision of keeping or dropping a predictor variable. In some cases, forward and backward selection resulted in 2 competing models. We used Akaike's Information Criterion (Boyce et al. 2002, Rushton et al. 2004) to identify the model that accounts for the most variation with the fewest variables. Additionally, we provide R -square Nagelkerke as a measure to compare the fit of habitat models built with different data sets (Pre-Alps, Alps).

Model evaluation.—For evaluating the models, we used measures based on a confusion matrix (Fielding and Bell 1997, Guisan and Zimmermann 2000, Boyce et al. 2002). A confusion matrix contains the predicted and observed presences and absences based on a fitted model. From this matrix, a large number of measures can be derived. We used Kappa statistics (Monserud and Leemans 1992) that measure the actual agreement minus the agreement expected by chance. Kappa takes values between zero and one (0.00–

0.05 = no agreement, 0.05–0.20 = very poor, 0.20–0.40 = poor, 0.40–0.55 = moderate, 0.55–0.70 = good, 0.70–0.85 = very good, 0.85–0.99 = excellent, 0.99–1 = perfect agreement). We used Kappa at the optimized threshold ($Kappa_{opt}$). To determine the optimized threshold, we calculated Kappa for all possible threshold values from 0.01 to 0.99. Because Kappa depends on a particular threshold, we also used the Receiver Operating Characteristic (ROC; Deleo 1993). The area under the ROC function (AUC) is usually taken to be an important index because it provides a single measure of overall accuracy that is not dependent upon a particular threshold (Fielding and Bell 1997, Boyce et al. 2002). The AUC can take values between zero and one. A value of 0.8 for the AUC, for example, means that for 80% of the time a random selection from the positive group will have a score greater than a random selection from the negative group.

We evaluated all 4 habitat models with 3 different approaches. First, we tested model performance on the data used for calibration (internal evaluation). Second, we partitioned the calibration data set randomly into 10 equal subsets and calculated AUC separately for each subset (10-fold cross-validation). In the results, we give mean and standard deviation of AUC for all models (AUC_{Mean} , SD). Further, we tested the 2 models calibrated with data from the Pre-Alps on data from the Alps and, inversely, the 2 Alpine models on data from the Pre-Alps (independent validation).

RESULTS

Plot Scale (Forest-Stand Mosaic)

The mean forested area per 5-km² plot was 340 ha in the Pre-Alps and 373 ha in the Central Alps. In the Pre-Alps, forest area did not differ between plot categories. In the Alps, forest area was higher in plots with capercaillie than in the plots of category extinct (404.1 ha and 298.4 ha, respectively; $U = 9.0$, $P = 0.001$) and the plots of the category stable had higher values than category declining (432.2 ha and 379.1 ha, respectively; $U = 12.0$, $P = 0.020$). Forest edge-length was higher in the Pre-Alps than in the Alps (44.6 km and 25.7 km, respectively; $U = 80.0$, $P < 0.001$). In the Pre-Alps, forest edge-length was higher in plots with capercaillie than in plots without capercaillie (50.0 km and 36.3 km, respectively; $U = 49.0$, $P = 0.036$).

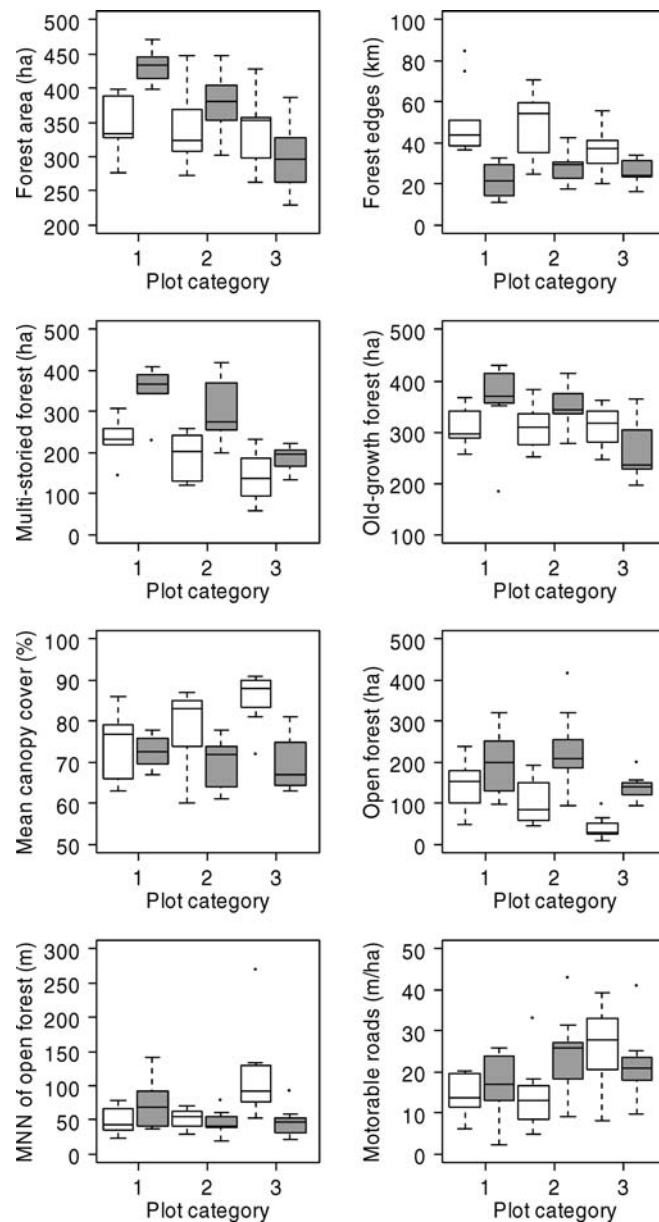


Figure 3. Relation of selected variables to capercaillie population status (based on inventories 1971, 1986, and 2001) at the scale of study plots of 5 km² in the Swiss Alps. These variables characterized the plots with respect to landscape composition (forest area, forest edges), forest structure (multi-storied forest, old-growth forest, mean canopy cover, open forest [total canopy cover between 25% and 75%], and mean nearest neighbor of patches of open forest [MNN of open forest]) and a surrogate for human disturbance (motorable roads). Plot categories represent different capercaillie population trends: stable (plot category 1), declining (plot category 2), and extinct (plot category 3). Separate clusters for Pre-Alps (white) and Alps (grey). Box plots show the median, quartiles, and outliers.

The proportion of multi-layered forest was lower in the Pre-Alps than in the Alps (188 ha and 285 ha, respectively; $U = 151.0$, $P = 0.001$). In both regions, however, plots with capercaillie contained more multi-layered forest than plots without capercaillie (Pre-Alps: $U = 35.0$, $P = 0.006$; Alps: $U = 3.0$, $P < 0.001$), with the category stable showing the highest values. The proportion of old-growth forest in relation to overall forest area did not differ between plot

Table 3. Accuracy of the 4 multivariate habitat models for capercaillie in the Swiss Alps, 1995–2003; for the models we used data from 1) the Pre-Alps and variables relating to forest structure only (PA_FS), 2) the Pre-Alps with forest and landscape variables (PA_FLS), 3) the Alps with forest-structure variables only (A_FS), and 4) the Alps with forest and landscape variables (A_FLS). We report Akaike's Information Criterion (AIC) and R^2_N for model fit, Kappa at the optimized threshold ($Kappa_{opt}$) and the area under the Receiver Operating Characteristic curve (AUC) for model validation, and mean AUC (AUC_{mean}) plus standard deviation for 10-fold cross-validation.

Model	Model fitting		Performance on calibration data	
	AIC	R^2_N	$Kappa_{opt}$	AUC
PA_FS	330.01	0.35	0.48	0.79
PA_FLS	229.86	0.66	0.68	0.92
A_FS	194.72	0.35	0.50	0.80
A_FLS	114.21	0.76	0.80	0.95
	Cross-validation (10-fold)		Validation on Pre-Alps, Alps	
	AUC_{Mean}	SD	$Kappa_{opt}$	AUC
PA_FS	0.78	0.08	0.11	0.38
PA_FLS	0.90	0.04	0.27	0.68
A_FS	0.80	0.08	<0.01	0.46
A_FLS	0.94	0.04	0.31	0.67

categories in both regions. In the Alps, however, the absolute amount of old-growth forest differed significantly between plot categories ($H = 7.98$, $P = 0.019$), with highest values in the category stable.

Generally, mean canopy cover was higher in the Pre-Alps than in the Alps (80.0% and 70.9%, respectively; $U = 134.0$, $P < 0.001$). In the Pre-Alps, the canopy was on average less dense in the capercaillie presence plots than in plots without capercaillie ($U = 27.5$, $P = 0.002$), and lowest values occurred in the category stable. We found no differences between plot categories in the Alps. As a consequence, in the Pre-Alps the proportion of open forest (total canopy cover between 25% and 75%) was higher in plots with capercaillie than in the plot category extinct ($U = 10.0$, $P < 0.001$). We found the same results with absolute values in the Alps ($U = 25.0$, $P = 0.028$), but we found no differences between plot categories when we considered the ratio of open forest to total forest area ($U = 51.0$, $P = 0.589$). Similarly, in the Pre-Alps patches of open forest were better connected in capercaillie plots than in plots where capercaillie was absent ($U = 10.0$, $P < 0.001$; Fig. 3).

The density of motorable roads was higher in plots without capercaillie than in the capercaillie plots in the Pre-Alps (26.7 m/ha and 14.3 m/ha, respectively; $U = 23.0$, $P = 0.001$). In the Alps, no such pattern could be found.

Scale of Grid Cells within Plots (Forest Stand)

The multivariate models built with only forest-structure variables explained capercaillie occurrence moderately well (Table 3). In the Pre-Alps, the model contained 3 variables relating to forest structure (i.e., total canopy cover at the 100-m scale, its squared term, and multistoried forests [Table 4]). The fourth variable, describing variation in canopy cover, indicates that edges between different forest

Table 4. Predictor variables selected by capercaillie habitat models calculated for 2 different regions (Pre-Alps, PA; Alps, A) and with 2 different variable sets (Forest structure only, FS; forest and landscape structure, FLS) using presence-absence data. We assessed forest structure from black-and-white aerial photographs from 1993–2000 with a ground resolution of 0.65 m. The landscape structure variables originated from geographic data with country-wide coverage (national data bases, state 2001). We report the type of response and the significance of the variables (*P*-value).

Variables ^a relating to forest structure only					
Pre-Alps (PA_FS)			Alps (A_FS)		
Variable	Response	<i>P</i> -value	Variable	Response	<i>P</i> -value
CC100	pos	<0.001	CCU	neg	0.002
CC100 ²	neg	<0.001	CCV100	neg	0.057
CCV100	pos	<0.001	VS100	pos	0.010
VS100	pos	0.010	FE100	neg	0.027
Forest and landscape structure variables ^a					
Pre-Alps (PA_FLS)			Alps (A_FLS)		
Variable	Response	<i>P</i> -value	Variable	Response	<i>P</i> -value
CC100	pos	<0.001	CCU	neg	0.002
CC100 ²	neg	0.001	CCV100	neg	0.003
CCV100	pos	0.001	TAVE	pos	0.037
TAVE	pos	0.001	TAVE ²	neg	0.024
TAVE ²	neg	0.001	TOP	pos	<0.001
TOP	pos	<0.001	SLOPE	neg	0.033
SLOPE	neg	0.038	PFOR	pos	0.002
MIRE	pos	0.003	SETTL	neg	0.053
SETTL	neg	0.001			

^a For variable definitions, see Tables 1 and 2.

patches increase habitat suitability. The model built with data from the Alps contained 3 variables defining features that are avoided by capercaillie in this region: understory cover, variation of canopy cover, and forest edges. These habitat relationships found for the Pre-Alps and the Alps, respectively, are contradictory to some degree. Consequently, both regional models (built with data from Pre-Alps and Alps, separately) failed to predict capercaillie occurrence when applied to the other area (AUC = 0.38, AUC = 0.46, respectively; see Table 3 for other accuracy measures).

The inclusion of landscape variables increased model performance (AUC) by 0.14 in the Pre-Alps and by 0.15 in the Alps, respectively (Table 3). In contrast to the models with only forest-structure variables, these models showed at least some degree of generality (AUC = 0.68, AUC = 0.67, respectively). The variables selected by these models are partly identical to those included in the forest-structure models (Table 4).

Application of Models to Entire Study Plots

We recorded a higher average proportion of cells with predicted presence in plots with capercaillie than in the plots without capercaillie (Pre-Alps: $U < 10.0$, $P < 0.001$; Alps: $U < 16.0$, $P < 0.004$; Fig. 4). We also tested differences for predicted presence as proportion of the available forest area per patch because forest area differed between plot categories in the Alps (see above). Relative predicted

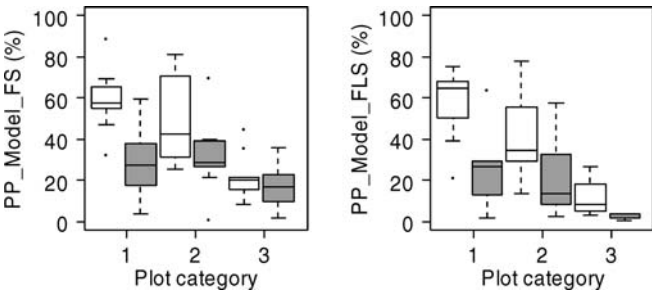


Figure 4. Relation of the proportion of presence cells per plot as predicted by the 4 habitat models (1995–2003) to capercaillie population trend (based on inventories 1971, 1986, and 2001) in the Swiss Alps. (left) Habitat models with forest-structure variables only (PP_Model_FS), (right) habitat models with forest and landscape structure variables (PP_Model_FLS). Plot categories represent different population trends: stable (plot category 1), declining (plot category 2), and extinct (plot category 3). Separate clusters for Pre-Alps (white) and Alps (grey). Box plots show the median, quartiles, and outliers.

presence area differed between presence and absence plots for the model with forest and landscape variables ($U = 9.0$, $P < 0.001$) but not for the model with forest-structure variables only ($U = 30.0$, $P = 0.065$). In the Pre-Alps, the proportion of forest area with predicted capercaillie presence was highest in the plot category stable (model with forest structure: 59.1%, model with forest and landscape structure: 56.6%, respectively), intermediate in the category declining (49.6% and 41.4%, respectively), and lowest in the plots without capercaillie (21.7% and 12.1%, respectively). In the Alps, the plot categories stable (59.8% and 63.3%, respectively) and declining (69.3% and 57.6%, respectively) had similar proportions, whereas the plot category extinct had the lowest proportions (36.2% and 3.9%).

DISCUSSION

Our results provide more evidence that both small-scale and landscape-scale characteristics of the habitat are significant predictors of capercaillie occurrence and abundance (cf. Storch 1997). At the scale of forest-stand mosaic, mean canopy cover, the proportion of open forest, and the proportion of multistoried forest differed significantly between plot categories. These forest-structure variables are known from literature to influence capercaillie habitat selection at the forest-stand scale; open forest or an intermediate canopy cover have been reported as important features for Scandinavian (e.g., Gjerde 1991a, Sjöberg 1996) and central European habitats (Storch 2002, Suchant 2002, Bollmann et al. 2005). A preference for multistoried forests as found here has been emphasized in older work (Eiberle 1976, Scherzinger 1976) and has been confirmed recently (Suter et al. 2002, Bollmann et al. 2005). This relationship, however, may reflect a preference for open forests that usually are characterized by multiple layers, whereas dense, unsuitable forest stands mostly show a uniform vertical structure (Suchant 2002). In boreal old-growth forests, however, the tree layer is not necessarily multistoried (Gjerde 1991a, Moss and Picozzi 1994).

At the scale of forest stands, forest-structure variables as

identified from aerial photographs could only partly explain winter occurrence of capercaillie. Three reasons may contribute to this: first, we lack information on 1) the extent of the field layer and its plant species composition, 2) the tree species composition, and 3) detailed stand structure data (e.g., Rolstad 1988, Schroth 1992, Storch 1993a, Bollmann et al. 2005). Using such variables would probably increase model fit. Second, capercaillie may partly use habitats that would be classified as unsuitable by expert opinion (Storch 1997); thus, it is likely that our presence data sets include capercaillie observations in unfavorable habitat, which would decrease model fit. Third, an unknown proportion of our absence cells may actually be used at least temporarily by capercaillie. In spite of these constraints, the proportion of grid cells predicted as presence (based on the habitat models with forest-structure variables only) is significantly higher in plots with capercaillie than in plots where capercaillie has disappeared, a finding in favor of model credibility.

Only the variable “proportion of multistoried stands” was common to the models of both regions. The possible advantage of diversely structured stands has been discussed above. Preference for an intermediate canopy cover has often been reported (e.g., Gjerde 1991a, Storch 2002, Suchant 2002). In our case, canopy cover was an important variable in the Pre-Alps but, surprisingly, not in the Alps. We attribute this to the fact that forest stands are generally open in the Alps, so that large proportions of potential capercaillie habitat occur in those areas where the species is actually absent (Fig. 3). Similarly, Åberg et al. (2000) concluded that difficulties in detecting habitat selection by animals occurred due to generally suitable composition and small variation of habitats within the study area.

Canopy-cover variation can be interpreted as a measure for the proportion of within-forest boundaries. This variable was a positive predictor in the Pre-Alps and a negative one in the Alps. Most often, a high density of boundaries within forests is seen as a positive habitat feature (De Franceschi and Bottazzo 1991, Schroth 1992), especially in areas where forests are generally dense and uniform due to intensive management (Suchant 2002) as is often the case in the Pre-Alps. However in the Alps, external edges between forest and farmland (as in the Alps) may have a negative effect because they allow higher numbers of generalist predators to intrude into the forest and, hence, to decrease the reproductive success of grouse species (Kurki and Linden 1995, Kurki et al. 2000). Additionally, large parts of Alpine forests are naturally open so that capercaillie do not depend on edge habitats. In the Pre-Alps, good habitat quality at the fringes of the frequent mires may compensate the negative influence of boundaries. Finally the high proportion of dense forest in the Pre-Alps may force capercaillie to use edge habitats.

Landscape variables used in earlier work explained much of the variation in capercaillie occurrence and improved habitat models at the forest-stand scale substantially. This lends support to Storch (2002), who concluded from her

results that landscape variables probably accounted for much of the unexplained variation. In our case, important variables were average temperature, topographic position, slope, and density of settlements for both regions. A similar set of landscape variables also explained capercaillie occurrence with a coarser definition of presence-absence (Graf et al. 2005, 2006).

Species-habitat relationships can differ markedly between ecologically different regions (Collins 1983). Unlike many previous studies, we included 2 ecologically differing regions and tested all habitat models on independent data from a spatially separated region. If we had selected Pre-alpine study areas only, our results would have been largely in agreement with most other studies from central European habitats (e.g., Schroth 1992, Storch 2002, Suchant 2002). However, our results at the scales of the forest stand and the stand mosaic for the central Alps revealed that environmental conditions and capercaillie-habitat relationships differ between the 2 regions. As a consequence, habitat models built in one region have limited predictive power in the other region (see, e.g., Graf et al. 2006). This confirms the importance of comparative studies between regions (Whittingham et al. 2003) and makes our cross-regional approach particularly valuable.

MANAGEMENT IMPLICATIONS

Conservation measures to halt the overall decline of capercaillie have rarely succeeded, probably because they were local in scope and neglected population processes operating at the landscape scale (Storch 2002). Conservation approaches would be more promising if they adopted a multi-scale habitat concept (Storch 1997). At an international or national level, large-scale distribution models (e.g., Graf et al. 2005) are an essential tool to identify priority areas with the local populations that need to be conserved (Mollet 2006). At the scale of forest-stand mosaics, our analyses can serve regional authorities or forest-management planners to define those parts of priority areas where suitable habitat is not available in the required proportion. Finally, forest-management planners can use detailed habitat models at the forest-stand scale (e.g., Bollmann et al. 2005) to identify the currently unsuitable stands that need management interventions to become more suitable. Our study also shows that these management schemes have to be adapted regionally, because species-habitat relationships may differ substantially between regions.

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